

SATURNA ISLAND EAST POINT, BC, Canada

WMO: 714730

Lat: 48.7832N

Lon: 123.0460W

Elev: 24

StdP: 101.03

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.7	-0.1	-10.1	1.6	0.4	-7.7	2.0	1.4	20.9	9.1	18.9	8.9	9.4	60	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	25.2	18.5	23.7	17.7	22.3	16.9	19.6	24.0	18.4	22.5	17.4	21.3	2.3	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.7	22.5	16.5	11.8	20.9	15.6	11.1	19.5	55.9	24.2	52.0	22.5	48.9	21.6	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	13.2	10.9	DB	-3.4	27.7	2.6	1.9	-5.3	29.1	-6.8	30.2	-8.2	31.3	-10.0	32.7	(4)
(5)			WB	-5.2	19.9	2.5	1.8	-7.0	21.2	-8.5	22.2	-9.9	23.2	-11.8	24.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	6.1	6.3	8.0	10.0	13.1	15.4	17.7	17.7	15.3	11.4	8.0	5.8
	DBStd	4.96	2.96	2.58	2.26	2.14	2.37	2.58	2.54	2.17	2.05	1.94	2.60	2.82
	HDD10.0	531	122	105	68	24	3	0	0	0	0	8	68	132
	HDD18.3	2633	378	338	321	249	163	96	42	39	92	215	311	389
	CDD10.0	991	2	1	6	25	98	161	239	238	160	52	8	2
	CDD18.3	51	0	0	0	0	1	7	22	18	3	0	0	0
	CDH23.3	171	0	0	0	0	2	29	87	50	2	0	0	0
	CDH26.7	17	0	0	0	0	0	2	12	2	0	0	0	0
Wind	WSAvg	4.3	5.3	5.1	4.8	4.0	3.4	3.5	3.5	3.4	3.1	3.9	5.5	6.0
Precipitation	PrecAvg	837	121	73	77	51	42	35	22	27	46	94	125	122
	PrecMax	1036	193	154	149	116	75	68	51	107	138	258	217	206
	PrecMin	588	60	13	24	2	0	3	0	3	4	10	46	46
	PrecStd	116	36	31	30	26	20	17	13	23	32	51	48	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	12.0	14.5	18.3	22.4	26.4	28.4	26.6	23.3	17.2	14.7	12.9
		MCWB	11.2	10.2	10.6	14.0	15.4	18.3	20.9	19.6	17.1	14.2	12.5	11.3
	2%	DB	11.6	10.7	12.8	15.9	20.0	23.7	25.5	24.7	21.2	15.9	13.0	11.3
		MCWB	10.3	9.0	10.0	11.9	14.3	17.1	18.8	18.7	16.2	13.5	11.6	9.8
	5%	DB	10.6	9.9	11.7	14.4	18.3	21.6	23.9	23.3	19.9	14.8	12.0	10.1
		MCWB	9.4	8.4	9.5	11.0	13.7	16.1	17.9	17.8	15.5	12.7	10.7	8.6
	10%	DB	9.7	9.2	10.9	13.2	16.9	19.7	22.3	21.9	18.6	13.9	11.2	9.2
		MCWB	8.5	7.8	8.9	10.3	12.9	15.1	17.0	17.0	14.9	12.1	9.9	7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	10.6	11.9	14.7	17.0	19.5	21.5	21.0	18.2	15.5	13.0	11.8
		MCDB	12.4	11.5	13.2	17.6	20.4	24.5	26.8	25.8	21.9	16.2	13.8	12.7
	2%	WB	10.7	9.5	10.8	12.6	15.4	17.9	19.9	19.4	17.1	14.2	11.8	10.0
		MCDB	11.2	10.4	11.9	15.1	18.6	22.6	24.5	23.6	20.1	15.1	12.7	11.1
	5%	WB	9.6	8.7	10.0	11.5	14.3	16.5	18.6	18.2	16.2	13.2	11.0	8.9
		MCDB	10.4	9.6	11.2	13.6	17.4	20.6	22.7	22.2	19.0	14.3	11.8	9.8
	10%	WB	8.7	8.1	9.2	10.6	13.3	15.4	17.4	17.2	15.5	12.4	10.2	8.1
		MCDB	9.5	8.9	10.6	12.7	16.3	19.2	21.4	21.1	18.0	13.5	11.0	9.0
Mean Daily Temperature Range	5% DB	MDBR	3.0	3.3	4.3	5.2	6.4	7.1	8.0	7.6	5.6	3.7	3.2	2.9
		MCDBR	3.7	3.8	5.2	6.5	8.2	9.6	9.8	9.3	7.2	4.5	4.0	3.6
		MCWBR	2.8	2.8	3.4	3.9	4.6	4.9	4.9	5.0	3.9	3.1	3.1	2.8
	5% WB	MCDBR	3.6	3.5	4.8	6.1	7.7	9.0	9.5	9.2	6.7	4.3	3.7	3.5
		MCWBR	2.9	2.8	3.5	4.0	4.7	5.0	5.4	5.3	4.0	3.2	3.1	3.0
Clear-Sky Solar Irradiance	taub	0.319	0.314	0.331	0.359	0.380	0.356	0.351	0.358	0.342	0.332	0.320	0.311	
	taud	2.517	2.558	2.526	2.438	2.393	2.476	2.520	2.504	2.548	2.559	2.529	2.499	
	Ebn at Noon	752	852	890	892	881	905	904	883	866	815	743	709	
	Edh at Noon	61	72	87	104	114	105	100	97	84	71	59	55	
All-Sky Solar Radiation	RadAvg	0.90	1.77	2.73	4.18	5.40	5.86	6.45	5.45	3.84	2.11	1.07	0.72	
	RadStd	0.13	0.25	0.34	0.41	0.44	0.61	0.50	0.38	0.32	0.26	0.18	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page